Anas Benbaki

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Final-year AI engineering student (Master's level). Available for a 6-month internship from February 2027.

Experience

Quantum Machine Learning Research Intern May 2026 – Aug. 2026
University of Calgary, Mitacs Globalink Research Internship *Calgary, Canada*
- Applied machine learning to the reconstruction of quantum electronic structure: predicting three-body correlation corrections from lower-order reduced density matrices (RDMs).
- Built the full Python pipeline end to end: quantum-chemistry calculations, dataset generation and model training.
- Compared gradient-boosted trees (XGBoost) with several multilayer-perceptron architectures, and identified a possible "locality ceiling" on how much of the correction lower-order information can predict.

Mathematics Tutor for AI Models (RLHF) Jan. 2025 – May 2025
Mercor AI *San Francisco, US (remote)*
- Evaluated and corrected large language model solutions to Mathematical Olympiad problems (algebra, combinatorics, number theory) for a reinforcement-learning-from-human-feedback training pipeline.
- Identified logical errors and unjustified steps in model-generated proofs, verified arguments and gave structured feedback on reasoning quality.

Full-Stack Developer Intern Jun. 2025 – Jul. 2025
Cap & Pro France *Bordeaux, France*
- Designed and deployed a hotline platform with an interactive voice response (IVR) system, integrated through telephony APIs, helping people with disabilities access employment support.

Projects

Reinforcement Learning Market-Making Agent *Python, PyTorch, C++, Binance API*
- Developed an Advantage Actor-Critic (A2C) deep reinforcement learning agent in PyTorch for autonomous market making.
- Built a limit order book simulator in C++ generating market microstructure data (bid-ask spread, order imbalance) for training and backtesting, then connected the agent to live Binance market data.

LLM and RL Agents for Automatic Program Optimisation *Python, LLVM, MAQAO, C++*
- Building an agent that selects and orders compiler optimisations (loop tiling, fusion, unrolling, vectorisation, data-layout changes) from LLVM and MAQAO profiling data.
- Comparing a compiler baseline, an LLM-driven agent, guided search and reinforcement learning, with measured run-time speed-up as the reward.

Hallucination in Image Captioning *Python, PyTorch*
- Comparing a CNN–LSTM captioning model with a CLIP + LLM pipeline, focused on when and why models describe content that is not in the image.

Conversational RAG Chatbot *Python, LangChain, Streamlit*
- Built and deployed a retrieval-augmented generation pipeline for question answering over PDF documents (chunking, embeddings, similarity search) – live demo.

Education

ENSEIRB-MATMECA (Bordeaux INP) 2024 – Expected Sept. 2027
Diplôme d'Ingénieur (Master's level) in Computer Science, Artificial Intelligence specialisation *Bordeaux, France*
- **Year 2 (2025–2026):** 15.35/20 (4.0/4.0 under ENSEIRB-MATMECA's official conversion), ranked **8th of 86**.
- **Relevant coursework:** Machine Learning and Deep Learning, Artificial Intelligence, Data Science, Probability and Statistics, Game Theory, Flow Algorithms and Combinatorics, Computability and Complexity, Cryptography.

Lycée d'Excellence de Benguerir (LYDEX) 2022 – 2024
Classes Préparatoires (CPGE), MP track (Mathematics & Physics) *Ben Guerir, Morocco*
- Two-year intensive programme in advanced mathematics and physics for the competitive entrance exams to French engineering schools.
- **Mathematics:** linear algebra, topology of normed vector spaces, real analysis, probability, differential equations.

Skills & Activities

Programming: Python, C++, C, SQL, Bash, Git, Linux
Machine learning: PyTorch, scikit-learn, NumPy, XGBoost, LangChain; deep learning, reinforcement learning, RLHF, LLMs, RAG
Mathematics: linear algebra, probability, statistics, real analysis, differential equations, graph theory, combinatorics
Languages: English (fluent, IELTS 8.0), French (fluent), Arabic (native)
Teaching: private mathematics tutor for university and high-school students (linear algebra, calculus, differential equations)
Mathematical Olympiad: IMO Team Selection training in high school; multiple national training camps run by Morocco's Ministry of Education and Math & Maroc
Teaching: private mathematics tutor for university and high-school students (linear algebra, calculus, differential equations)